Amazon Sales Analytics Using Data Science and Power BI

Problem

Order, customer, product and payment information is distributed across separate tables. Without consistent preparation, relationships and metric definitions, reporting can produce misleading counts and inflated sales totals. This project brings those sources into an analytical workflow and an interactive Power BI report.

Expected Outcome

A repeatable workflow for profiling and preparing sales data, building a relational analytical layer, exploring purchasing patterns and presenting sales, customer, product and transaction insights through five dashboard pages.

The analysis covers 10,000 orders, 55,136 units and 6,340 purchasing customers. The gross sales proxy is 13,987,259.77, calculated from quantity and listed product price.

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ABSTRACT

This project develops an Amazon sales analytics workflow using Python, SQL and Power BI. Nine CSV source tables cover customers, orders, products, delivery partners, transaction attempts, ratings, returns, subscription details and subscription plans. Order dates span 1 January 2023 to 31 December 2024.

Python and Pandas support profiling, duplicate removal, missing value treatment, type conversion, relationship validation and exploratory summaries. SQL defines staging tables and a typed relational analytical layer. Power BI presents Introduction, Sales Overview, Customer Insights, Product Performance and Transaction Analysis pages.

The analysis identifies 10,000 orders, 55,136 units and 6,340 purchasing customers. The gross sales proxy is 13,987,259.77 and the average order value proxy is 1,398.73. The 2024 sales proxy exceeds 2023 by 4.10%. Electronics contributes the largest category sales proxy, and 2,633 purchasers place more than one order during the observed period.

Payment analysis covers 10,000 attempts linked to 6,248 distinct orders. The successful attempt rate is 49.76%, which differs from a final order payment outcome. Preparation also identifies 1,177 reversed subscription intervals, demonstrating the need for business validity checks after successful date parsing.

The project demonstrates descriptive analytics and interactive reporting. The sales measure uses listed prices rather than settled revenue, and the findings describe this project dataset rather than official Amazon business performance. No predictive model is trained or evaluated.

Keywords

Sales analytics, Python, Pandas, SQL, Power BI, exploratory data analysis, data quality, customer frequency, dashboard.

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CHAPTER 1 INTRODUCTION

1.1 Background

Ecommerce analysis needs to explain what customers purchase, how demand changes over time and how supporting payment and service records relate to orders. A dashboard becomes useful when these questions are supported by consistent data preparation and clear measures.

1.2 Problem Statement

The task is to transform separate sales related files into validated analytical tables and an interactive report. The solution must preserve each source table's grain so that repeated transactions, ratings or returns do not multiply order totals.

1.3 Objectives

- Profile all nine source tables and document quality issues.
- Prepare identifiers, dates, numeric values and missing categories.
- Validate dimension keys and child to parent relationships.
- Analyze monthly trends, category demand, geography and customer frequency.
- Distinguish payment attempt outcomes from distinct order outcomes.
- Present findings using KPI cards, charts, detail tables and filters.

1.4 Scope

The project uses a fixed dataset for 2023 and 2024. It covers Python preparation and EDA, SQL analytical design and Power BI presentation. Live Amazon integration, payment settlement, profitability, inventory optimization and forecasting are outside its scope.

1.5 Expected Contribution

The report connects preparation choices to interpretable business measures. It enables users to explore date and category selections while retaining the definitions needed to compare results correctly.

PROJECT WORKFLOW

1.6 Workflow Stages

Stage	Main activity	Output
Collection	Load nine CSV tables	Raw data frames
Profiling	Inspect shape missingness and duplicates	Source inventory
Preparation	Standardize fields and remove duplicates	Clean tables and flags
Validation	Check keys relationships and ranges	Integrity checks
Analysis	Aggregate time category state and customer	EDA summaries
SQL modeling	Define staging dimensions and facts	Relational layer
Power BI reporting	Configure measures visuals and filters	Five page report

Python and SQL implement parallel preparation paths. The notebook does not export cleaned CSVs or load MySQL directly. Staging import and downstream refresh are separate operational steps that require a documented execution order.

1.7 Analytical Questions

- How do orders, units and the sales proxy change over time?
- Which categories and states contribute the largest shares?
- How many customers purchase repeatedly within the period?
- How widely is demand distributed across the catalog?
- What do payment coverage and overlapping statuses reveal?

1.8 Chapter Summary

The workflow combines data quality controls, descriptive analysis and interactive reporting. Its central design principle is to calculate each metric at the correct grain and explain its denominator.

CHAPTER 2 TECHNOLOGY REVIEW

2.1 Python and Pandas

The notebook uses Python and Pandas to load CSVs, profile missingness, remove duplicate rows, convert types, validate joins and produce grouped summaries. Raw and clean dictionaries preserve the imported data alongside transformed working copies.

2.2 Jupyter Notebook

The notebook contains 25 cells, including 24 code cells. Its sequential structure keeps data inspection, preparation and exploratory outputs together. Parameterized paths and explicit assertions would improve portability and repeatability.

2.3 SQL

The MySQL oriented script separates text staging tables from typed dimensions and facts. Primary keys identify records, foreign keys express relationships and decimal fields give monetary storage an explicit precision. Reporting queries summarize sales, payments, returns, ratings and subscriptions.

2.4 Power BI and DAX

Power BI provides the report canvas, KPI cards, slicers, charts and detail tables. DAX supports measures that respond to filter context. Distinct order and customer counts and ratios require particular care because their denominators change with the selected population.

2.5 Exploratory Data Analysis

EDA examines grouped summaries and distributions before drawing conclusions. This project focuses on sales over time, category and state shares, product demand, customer frequency and payment method outcomes. These comparisons are descriptive rather than causal.

2.6 Technology Selection

Technology	Role
Python and Pandas	Preparation profiling validation and aggregation
Jupyter Notebook	Sequential analytical record
SQL	Typed relational schema and reusable summaries
Power BI	Interactive report and visual exploration

CHAPTER 3 SYSTEM DESIGN

3.1 System Overview

Orders form the central analytical grain. Customer, product and delivery dimensions describe them. Transaction, rating and return records retain their separate event keys. Subscriptions form a customer to plan branch, and Power BI adds a calendar dimension.

3.2 Functional Requirements

- Load and profile the nine sources.
- Preserve record identifiers and check duplicate keys.
- Apply documented missing value and conversion rules.
- Calculate order level sales and customer frequency measures.
- Support time category product customer and payment analysis.
- Provide usable report navigation and filters.

3.3 Non-Functional Requirements

Requirement	Design response
Accuracy	Validate join cardinality and reconcile headline totals
Usability	Lead with KPI cards and provide visual detail
Maintainability	Separate preparation modeling and presentation
Traceability	Preserve raw values record keys and exception flags
Privacy	Do not reproduce customer addresses or mobile numbers

3.4 Architecture

CSV sources feed Python profiling, preparation and EDA. A separate source import feeds SQL staging and typed reporting tables. Power BI presents the analytical model. Common definitions and reconciliation totals connect these paths and prevent silent drift.

Source Inventory and Analytical Scope

The project begins with 62,548 rows across nine CSV files. Removing 209 exact duplicate dimension rows leaves 62,339 records across those files. Order records retain their original 10,000 row grain.

Source table	Raw rows	Clean rows	Grain
customer	10,184	10,000	Customer ID
delivery	325	300	Delivery partner ID
orders	10,000	10,000	Order ID
products	10,000	10,000	Product ID
ratings	10,000	10,000	Rating record ID
returns	2,000	2,000	Return record ID
subscription details	10,036	10,036	Subscription record ID
subscription plan	3	3	Plan ID
transactions	10,000	10,000	Transaction ID

3.5 Data Flow

Filter selections define the reporting population. Order measures summarize Fact_Orders; payment measures summarize Fact_Transactions; subscription interval measures use valid records. Child records are aggregated before any comparison with order totals.

Data Model and Relationships

The SQL script creates a staging layer with text fields and a typed analytical layer with four dimension tables and five fact tables. Power BI adds a date dimension. This is an order centered relational model with a separate customer subscription branch.

Parent	Child	Relationship
Customer Product Delivery	Orders	One to many
Orders	Transactions Ratings Returns	One to many
Customer and Plan	Subscriptions	One to many

The diagram summarizes the intended relationships in SQL and the date table visible in PBIX metadata. It does not assert that every Power BI relationship has the same active state or filter direction.

Grain and integrity

Fact_Orders uses Or_ID as its primary key. Transaction, rating and return tables have their own record keys and reference orders; these are not guaranteed to contain one row per order. Fact_Subscriptions references customer and plan dimensions and retains the invalid date interval flag.

```
INSERT INTO amazon_sales.fact_orders (...)
SELECT o.Or_ID, ...,
       CAST(NULLIF(TRIM(o.Qty), '') AS UNSIGNED),
       ..., p.Price,
       CAST(NULLIF(TRIM(o.Qty), '') AS UNSIGNED) * p.Price
FROM amazon_sales.stg_orders AS o
JOIN amazon_sales.dim_product AS p ON p.P_ID = o.P_ID;
```

Primary keys, foreign keys and indexes on order dates and child order IDs give the schema a clear integrity structure. DECIMAL fields make the storage precision of prices and sales proxies explicit. Blank text is handled with TRIM, NULLIF and COALESCE in several transformations.

3.6 Design Tradeoffs

Keeping each fact at its own grain prevents order value duplication and makes status overlap visible. A single flattened table would be simpler to browse but could misstate totals when several child records refer to the same order.

CHAPTER 4 IMPLEMENTATION

4.1 Development Approach

The implementation uses the project notebook for data preparation and exploratory summaries, the SQL script for analytical schema and queries, and Power BI for presentation. The following steps describe the actual project workflow and its controls.

The notebook separates raw and clean dictionaries, which keeps the original imported frames available for comparison. Loading identifiers as strings protects their prefixed formats and avoids accidental numeric treatment.

A controlled cleaning sequence

Profiling checks shape, column names, missing values and exact duplicates. Repeated customer and delivery IDs are then tested for conflicting attributes before deduplication. Both conflict checks return zero, supporting the decision to remove identical copies rather than choose between competing records.

```
raw = {
    name: pd.read_csv(DATA / filename, dtype="string")
    for name, filename in files.items()
}
clean["customer"] = clean["customer"].drop_duplicates()
```

The remaining entity keys are checked for duplication, and eight child to parent key comparisons return zero orphan IDs. Numeric and date conversion failures are separately identified using the original nonmissing values, so source blanks are not misclassified as conversion errors.

Joins and metric construction

The order enrichment joins product and customer attributes with validate="many_to_one". This is a useful guard against accidentally multiplying order rows when a dimension key is duplicated. The enriched frame remains at 10,000 distinct orders and has no missing joined product prices.

```
orders_eda["Gross_Sales_Proxy"] = (
    orders_eda["Qty"] * orders_eda["Price"]
)
orders_eda["Month"] = (
    orders_eda["Order_Date"].dt.to_period("M").dt.to_timestamp()
)
```

Data Cleaning and Validation

The main structural checks pass after deduplication: entity keys are unique, the eight tested relationships have no orphan IDs, and tested numeric and date conversions have zero failures. Valid parsing does not guarantee valid business chronology, as the subscription results demonstrate.

Issue	Observed count	Current treatment
Exact customer duplicates	184 excess rows	Remove identical rows; retain 10,000 unique customers
Exact delivery duplicates	25 excess rows	Remove identical rows; retain 300 partners
Missing coupon	5,011 orders	Label No Coupon; verify that blank truly means no coupon
Missing product name	14 products	Label Unknown Product; keep Product ID
Missing payment method	33 attempts	Label Unknown rather than discard
Missing ratings	22 product; 29 service; 20 partner	Keep missing numeric values; exclude from averages
Reversed subscription dates	1,177 of 10,036	Flag and exclude from duration and active interval summaries

What the checks support

Customer ages fall within the tested 18 to 100 range. Prices and quantities are positive, and nonmissing product and service ratings fall between 1 and 5. The notebook preserves missing ratings rather than replacing them with zeros, avoiding an artificial reduction in average satisfaction.

Checks still needed before operational use

Test whether returns precede their related orders and whether subscription periods overlap for the same customer. Collect rating event timestamps to enable a similar chronology check. Validate categorical values and required text after trimming whitespace. A blank coupon label is a semantic assumption, so distinguish unavailable coupon information if blanks can represent both meanings.

Keep exception records available for investigation. The 1,177 reversed subscription intervals represent 11.73% of subscription rows; silently deleting them would hide a material coverage problem.

4.2 Validation Outcome

Successful parsing and referential integrity establish a consistent technical structure. Business checks remain necessary for interval validity, event chronology and the meaning of missing categories.

Metric Construction and SQL Reconciliation

The analytical SQL mirrors the notebook in its month, category, state, payment, return, rating and subscription summaries. The following full period values were recomputed from the CSVs; the matching SQL logic and dashboard presentation were inspected as separate evidence.

Metric	Recomputed value	Interpretation
Orders	10,000	Distinct Or_ID; one row per order in this dataset
Units	55,136	Sum of Qty
Gross sales proxy	13,987,259.77	Sum of quantity times listed product price
Average order value proxy	1,398.73	Gross sales proxy divided by orders
Purchasing customers	6,340	Distinct customers with an order
Repeat purchasing customers	2,633	Purchasers with more than one order in context
Repeat customer rate	41.53%	Repeat purchasers divided by purchasers
Orders per purchasing customer	1.5773	Orders divided by purchasing customers

Avoid multiplying amounts across facts

Joining orders directly to transaction attempts before summing sales proxies would repeat an order amount for every related attempt. Aggregate attempts to the required order grain first, or keep order and transaction measures separate. Distinct order counts across payment methods or statuses are also nonadditive because an order can occur in multiple groups.

No discount amounts, refund values, tax fields or settled payment amounts are supplied. Neither the SQL proxy nor its dashboard card should be described as net revenue or realized cash.

4.3 Verification Method

The notebook was rerun against the source CSV collection. SQL expressions were inspected using the same metric definitions; live MySQL execution is not claimed. Exact values provide a common reference for checking rounded dashboard cards.

Power BI Dashboard Implementation

The report uses a consistent navy and orange design, a 1920 by 1080 canvas and five named pages. It contains 54 visual containers in total. The three analysis pages each have four KPI cards, a date slicer, a category slicer and supporting charts or detail tables.

Page	Visuals	Main purpose
Introduction	8	Project introduction and monthly sales overview
Sales Overview	10	Monthly category and state order performance
Customer Insights	12	Purchasing frequency and customer distribution
Product Performance	12	Category and individual product demand
Transaction Analysis	12	Payment attempt volumes and outcomes

Definitions confirmed in report metadata

Sales cards bind to the Fact_Orders measures Orders, Units, Gross Sales Proxy and Average Order Value Proxy. Customer cards bind to four Dim_Customer measures. Customer age groups and monthly category fields are referenced explicitly by the visual queries.

Transaction cards use a count of nonblank transaction IDs. Successful and failed cards apply explicit Tran_Status filters; the related order card uses a distinct order count. The monthly transaction visual groups by Dim_Date Year-Month, so its time axis is the related order date rather than a separate payment event timestamp.

Interpretation and usability

The purchaser card displays 6K even though the exact total is 6,340. Likewise, 13.99M and 1.40K are rounded sales and average order value proxies. Provide exact values in tooltips or supporting tables when the report is used for reconciliation.

Monthly distinct purchasers should not be summed to obtain the two year customer total. Repeat rate is a selected context frequency measure, not a cohort retention rate. Product IDs in the detail table distinguish different products with the same display name.

The model diagram lists ten tables, but visual metadata alone does not establish all DAX expressions or relationship behaviors. Export a PBIP or TMDL model definition for a complete semantic review and automated regression checks.

Measure Definitions

4.4 Order Measures

The following DAX illustrates the order measure definitions for reproducibility. It is a documented implementation pattern; the embedded semantic model should be exported to verify its complete formulas and relationship behavior.

```
Orders = DISTINCTCOUNT(Fact_Orders[Or_ID])
Units = SUM(Fact_Orders[Qty])
Gross Sales Proxy = SUM(Fact_Orders[Gross_Sales_Proxy])
Average Order Value Proxy =
    DIVIDE([Gross Sales Proxy], [Orders])
```

4.5 Customer Frequency

Purchasing customers are distinct customers with an order in the selected context. Repeat purchasers have more than one distinct order in that context. The repeat rate divides repeat purchasers by purchasers, while orders per purchaser divides distinct orders by purchasers.

4.6 Payment Measures

Attempt count uses transaction records. Successful attempts apply the Successful status filter. Related orders use distinct order identifiers. An order can appear under multiple methods and statuses, so grouped distinct order counts cannot be added to obtain an overall distinct count.

4.7 Date Context and Precision

Transaction trends use related order dates because a separate payment event timestamp is unavailable. Rounded display units make cards readable, but exact values are needed for reconciliation. Monetary values are presented without an assumed currency symbol because the analytical source fields do not establish a currency.

4.8 Implementation Summary

The implementation preserves identifiers, validates joins and calculates clearly defined metrics. Parameterized paths, controlled staging loads, automated assertions and a documented source of transformation ownership are the next steps toward a repeatable operational refresh.

CHAPTER 5 RESULTS AND DASHBOARD INSIGHTS

5.1 Dataset Results

The full period analysis covers 1 January 2023 through 31 December 2024. Headline values were recomputed from the source tables through the notebook workflow.

Indicator	Result
Orders	10,000
Units	55,136
Gross sales proxy	13,987,259.77
Average order value proxy	1,398.73
Purchasing customers	6,340
Repeat purchasers	2,633
Repeat customer rate	41.53%
Purchased product IDs	6,266

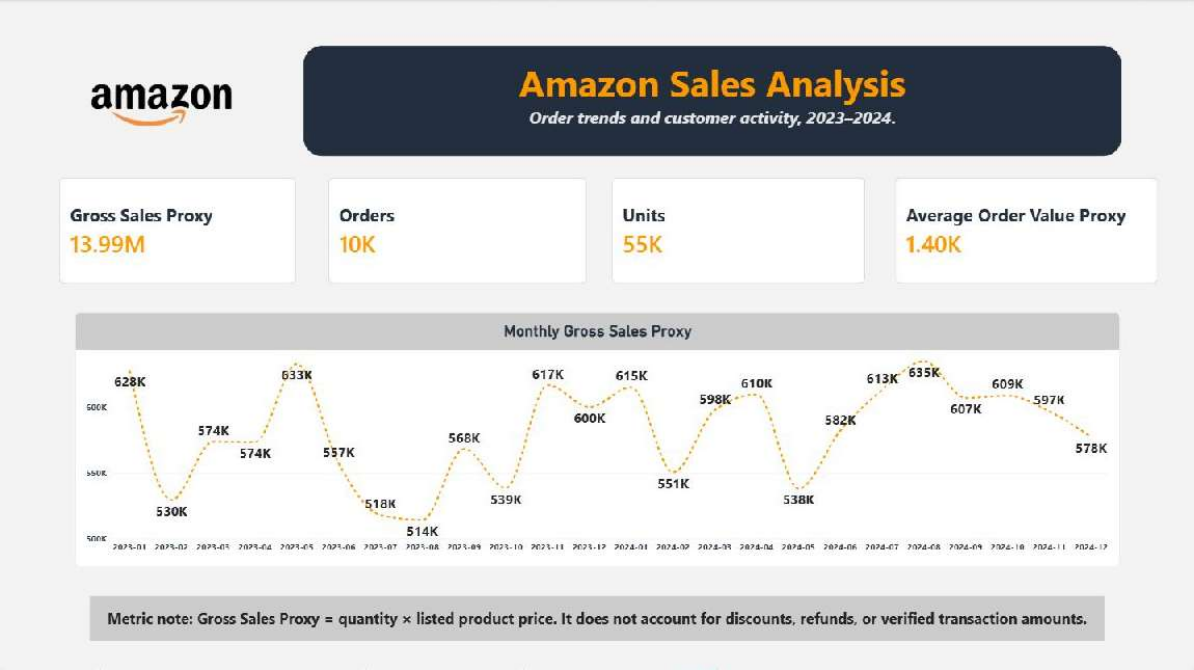


Figure 1 Introduction page showing headline indicators and the monthly trend

The landing page establishes the project context. The remaining pages provide focused views of sales, customers, products and payment attempts.

Sales Overview Results

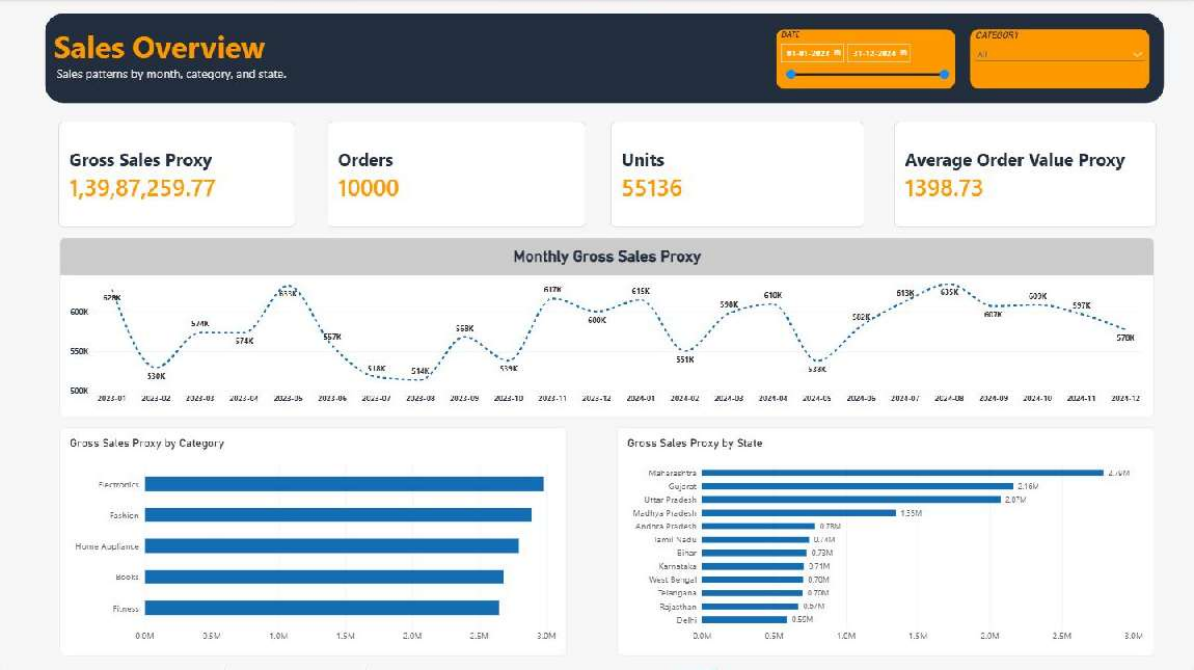


Figure 2 Sales Overview with the complete 2023 to 2024 date range and all categories.

Year	Orders	Units	Sales proxy
2023	4,977	27,205	6,852,986.53
2024	5,023	27,931	7,134,273.24

The sales proxy rises from 6,852,986.53 in 2023 to 7,134,273.24 in 2024, an increase of 4.10%. Orders grow by 0.92% and units by 2.67%. This is consistent with a contribution from basket quantity and price or product mix, but the aggregates do not isolate those effects.

August 2024 has the highest monthly sales proxy at 635,178.53. August 2023 is the lowest at 514,171.31. Only two annual cycles are available, so the monthly pattern is descriptive evidence rather than a reliable seasonal forecast.

Geographic concentration

Maharashtra leads with 2,788,692.40, followed by Gujarat at 2,160,971.01 and Uttar Pradesh at 2,073,340.30. Together they contribute 50.21% of the sales proxy. This concentration supports a closer look at regional service capacity, subject to validating the dataset population and business relevance.

Customer Insights Results

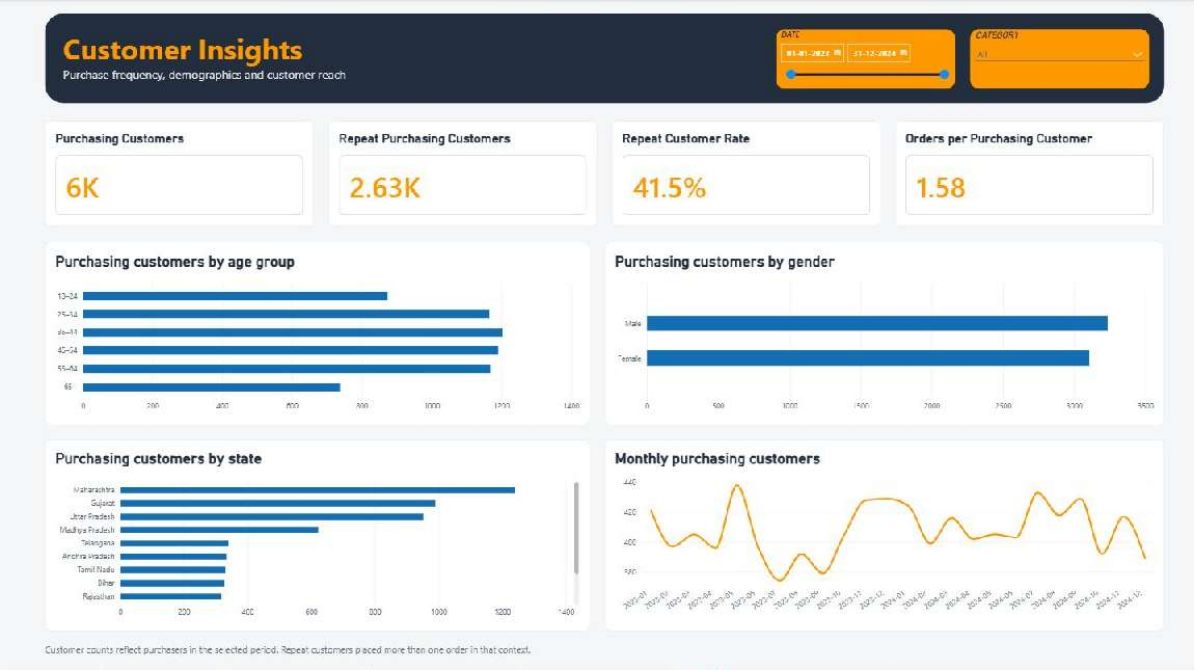


Figure 3 Customer Insights showing frequency cards demographic splits state reach and monthly purchasers.

Measure	Exact result
Purchasing customers	6,340 of 10,000 customer records
Repeat purchasers	2,633; 41.53% of purchasers
Orders per purchaser	1.5773
Male and female purchasers	3,234 and 3,106

The remaining 3,707 purchasers have exactly one order. Repeat purchasers account for 6,293 of the 10,000 orders, or 62.93%. This makes purchase frequency a useful segmentation variable, although it does not establish customer lifetime value or future retention.

The largest age group by purchasers is 35 to 44 with 1,204 people. The adjacent 45 to 54 and 55 to 64 groups contain 1,190 and 1,170 purchasers. These counts are similar; targeting decisions should also consider the underlying customer population, observed order value and sample coverage.

Maharashtra has 1,239 purchasing customers, Gujarat 989 and Uttar Pradesh 951. Monthly purchaser counts overlap because the same customer can appear in more than one month. A proper retention analysis would define an acquisition cohort, a follow up window and repeat purchase timing.

Product Performance Results

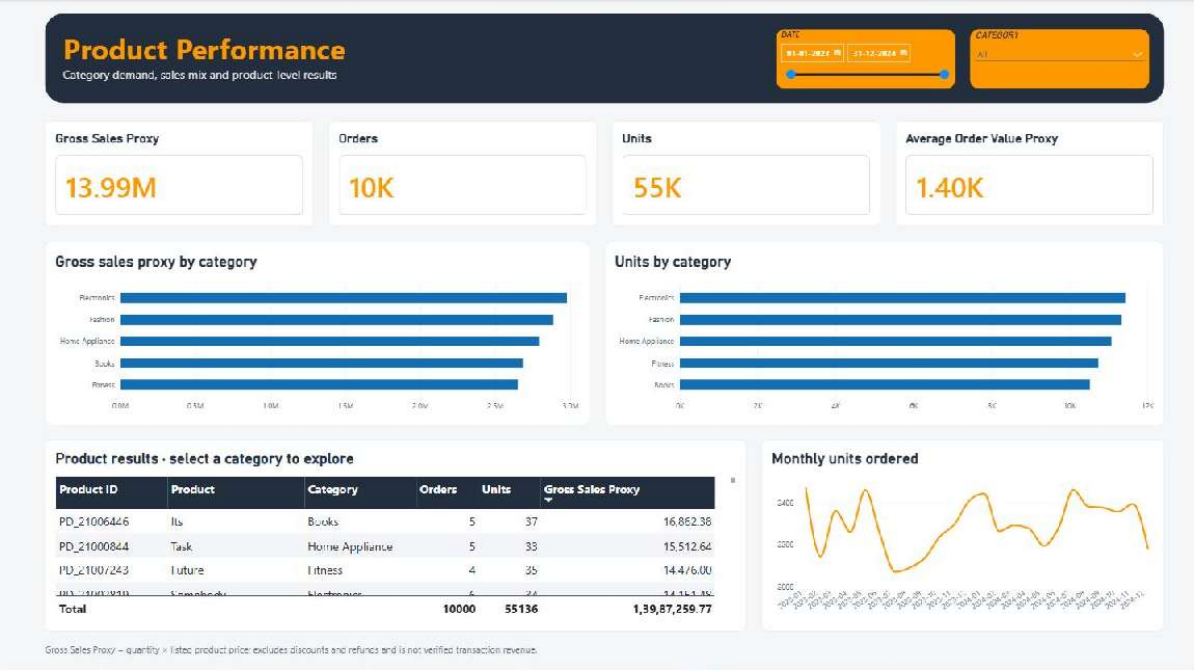


Figure 4 Product Performance with category comparisons product IDs and monthly units ordered.

Category	Orders	Units	Sales proxy
Electronics	2,055	11,450	2,977,529.58
Fashion	2,060	11,344	2,887,254.62
Home Appliance	1,999	11,077	2,790,930.93
Books	1,923	10,524	2,682,161.67
Fitness	1,963	10,741	2,649,382.97

Electronics has the largest sales proxy at 2,977,529.58 and the highest unit total at 11,450. Fashion has the most orders at 2,060, slightly above Electronics at 2,055. Category shares are fairly balanced, ranging from 18.9% for Fitness to 21.3% for Electronics.

Only 6,266 of the 10,000 listed product IDs appear in orders. The highest proxy product is PD_21006446, named Its, with 37 units across five orders and 16,862.38 in sales proxy. Different product IDs can share that name, so ranking by name alone would merge distinct items.

The top ten individual product IDs contribute only 1.01% of the overall sales proxy. The evidence describes a dispersed catalog rather than dominance by a small number of products. Margin, inventory availability, discount cost and return severity are needed before translating volume rankings into assortment actions.

Transaction Analysis Results

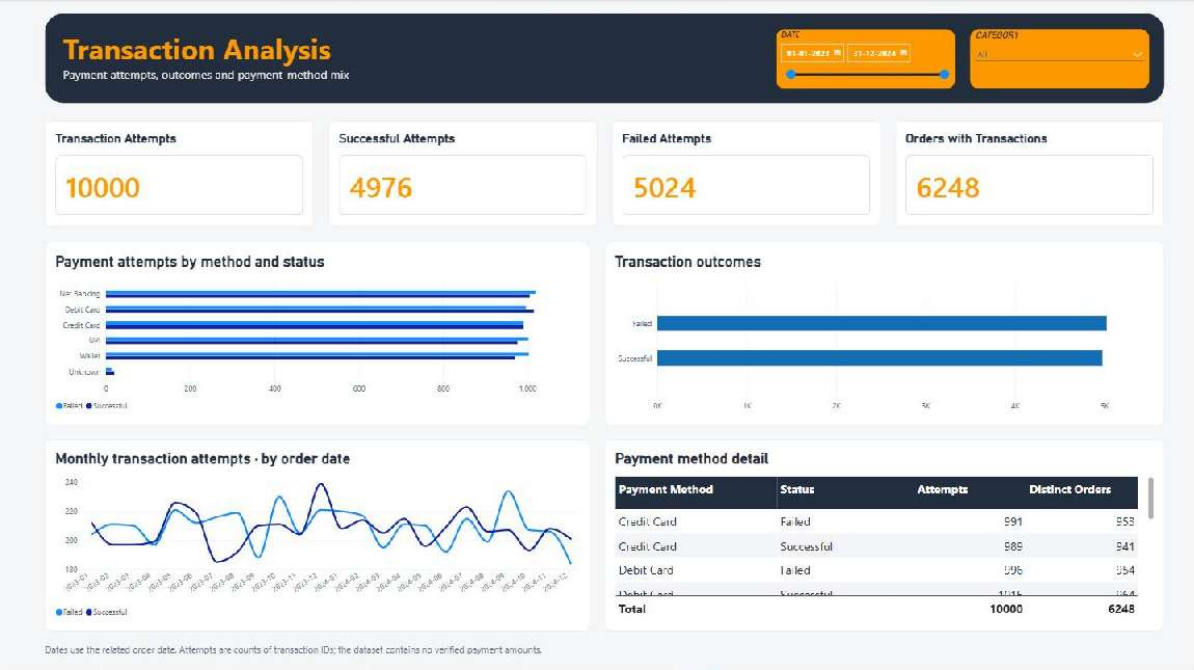


Figure 5 Transaction Analysis. Dates represent related order dates rather than payment event timestamps.

Payment method	Attempts	Successful	Success rate
Credit Card	1,980	989	49.95%
Debit Card	2,011	1,015	50.47%
Net Banking	2,024	1,005	49.65%
UPI	1,978	976	49.34%
Unknown	33	20	60.61%
Wallet	1,974	971	49.19%

There are 4,976 successful and 5,024 failed attempts, giving a 49.76% attempt success rate. The 10,000 attempts refer to 6,248 orders; 3,752 order IDs have no transaction record. This coverage gap is not evidence that those orders were unpaid.

Among represented orders, 2,632 have multiple attempts and 1,534 have both successful and failed records. A total of 3,854 orders have at least one successful attempt. Status counts therefore cannot be interpreted as mutually exclusive final order outcomes.

The five named methods have success rates close to one another. The Unknown group contains only 33 attempts, so its 60.61% rate is not a sound basis for comparing payment methods. Add event timestamps, failure codes and settlement references before diagnosing payment processing causes.

Returns Ratings and Service Results

The supporting fact tables add useful service evidence that is not yet surfaced on a dedicated dashboard page. Return records and rating responses must be analyzed at their own grains, with order level summaries used when the question concerns affected orders.

Return and refund incidence

Record type	Reason	Records
Refund	Late Delivery	352
Refund	Defective Product	342
Refund	Wrong Item Shipped	337
Return	Wrong Item Shipped	331
Return	Defective Product	328
Return	Late Delivery	310

The 2,000 return or refund records affect 1,820 distinct orders, an order incidence of 18.2%. There are 1,031 refund records and 969 return records. The 180 records above the distinct order total show why record counts and affected order counts must be reported separately.

Late Delivery, Defective Product and Wrong Item Shipped contribute 662, 670 and 668 records respectively. These categories are nearly even. A reason field supports descriptive triage, but it does not establish root cause, financial loss or service level compliance.

Ratings and response coverage

Rating type	Responses	Mean	Median
Product	9,978	2.99	3.00
Delivery or service	9,971	3.01	3.00

Missing values are excluded from averages. Product responses are missing for 22 records and service responses for 29. Separate partner ratings are missing for 20 of the 300 deduplicated delivery partners; they are a different measure and should not be blended with order service scores.

Recommended extension

Add a service page showing distinct affected orders, reason mix, response counts and distributions alongside averages. Validate return chronology against the order date and introduce refund amounts and processing timestamps before measuring financial impact or resolution time.

Subscription Results

Subscription analysis requires a valid start and end interval. The notebook flags reversed intervals and excludes them from duration and active snapshot calculations. SQL preserves the same flag in Fact_Subscriptions and filters it in the final reporting query.

Validation result	Count	Share of all rows
All subscription records	10,036	100.00%
Invalid reversed intervals	1,177	11.73%
Valid intervals retained	8,859	88.27%

Valid subscriptions by plan

Plan	Records	Customers	Median days	Active records
Prime Basic	2,950	2,530	419.0	1,714
Prime Standard	3,013	2,604	420.0	1,697
Prime Premium	2,896	2,509	416.5	1,625

The reporting snapshot is 31 December 2024, the latest order date. Valid intervals include 5,036 active subscription records at that snapshot, representing 3,960 distinct customers across all plans. Plan level customer counts can overlap and must not be added to obtain a unique customer total.

Prime Standard has the largest valid record count at 3,013, followed by Prime Basic at 2,950 and Prime Premium at 2,896. Median duration is about 417 to 420 days. These durations describe the recorded intervals; they are not evidence of contractual plan terms, billing cycles or customer renewal behavior.

Treat date validity as a release gate

Investigate the 1,177 rejected intervals with the source owner rather than swapping start and end dates automatically. Invalid records are spread across all three plans: 410 Basic, 380 Standard and 387 Premium. Preserve the original fields and an exception reason so corrections remain traceable.

The SQL median query uses ranked durations and averages the middle one or two values within each plan. Its fixed snapshot matches the notebook for this dataset, but should become a parameter when the reporting period changes. Add overlap rules and a distinct active customer measure before reporting subscriber penetration.

Objective Validation and Key Findings

5.2 Objective Validation

Objective	Evidence
Prepare sources	209 exact dimension duplicates removed
Validate relationships	Eight key checks with zero orphan identifiers
Reconcile sales	10,000 orders and 55,136 units
Study frequency	6,340 purchasers and 2,633 repeat purchasers
Inspect payments	Attempt coverage and overlapping statuses quantified
Present analysis	Five dashboard pages with source screenshots

5.3 Key Findings

- Sales proxy grows 4.10% year over year while orders grow 0.92%.
- Electronics leads category sales proxy; all five category shares remain close.
- Repeat purchasers contribute 62.93% of observed orders.
- Maharashtra Gujarat and Uttar Pradesh contribute 50.21% of the sales proxy.
- Successful attempt rate is 49.76% and is not a final order payment rate.
- Reversed intervals affect 11.73% of subscription records.

5.4 Interpretation

These comparisons describe the supplied project population. They do not establish causal drivers, profit, settled revenue or official Amazon performance. Two annual cycles support trend description but do not prove seasonality. Operational decisions require confirmation of data coverage and business definitions.

5.5 Chapter Summary

The report exposes demand and purchasing patterns while retaining the distinctions needed for reliable interpretation. Data quality, event coverage and consistent filter context remain central to further analysis.

CHAPTER 6 CONCLUSION

6.1 Conclusion

The project demonstrates a descriptive analytics workflow across source profiling, preparation, relational design, exploratory analysis and Power BI presentation. It organizes 10,000 orders into interpretable sales, customer, product and transaction views.

Its principal outcome is a consistent set of measures: 55,136 units, a sales proxy of 13,987,259.77 and 6,340 purchasing customers. Supporting analysis explains repeat purchasing, dispersed product demand and the differences between payment attempt records and order outcomes.

6.2 Project Achievements

- Established source inventories and documented cleaning choices.
- Preserved order grain through validated dimension joins.
- Calculated KPIs with explicit populations and denominators.
- Presented the analysis through five Power BI pages.
- Identified material subscription validity and payment coverage issues.

6.3 Limitations

The fixed two year dataset does not establish population representativeness. Listed prices do not capture historical transaction prices, realized discounts, tax, refunds or settlement. Payment records lack event timestamps and final settlement evidence. Complete semantic auditing requires exported Power BI measure and relationship definitions.

6.4 Design Tradeoffs

Separate tools support understandable analysis but can introduce differences between Python and SQL transformations. Common definitions, controlled execution and reconciliation checks reduce this risk. Rounded cards aid quick reading, while exact values remain necessary for verification.

REFERENCES

Project Sources

1. Amazon_sales.ipynb. Project notebook covering source loading, profiling, preparation, validation and descriptive analysis.

2. Amazon.sql. Project SQL script defining staging tables, dimensions, facts and reporting queries.

3. Amazon_Sales_Dashboard.pbix. Project Power BI report with Introduction, Sales Overview, Customer Insights, Product Performance and Transaction Analysis pages.

4. Source CSV collection in D:\amazon\Import Data. Nine tables covering customers, delivery partners, orders, products, ratings, returns, subscription details, plans and transactions.

5. Dashboard screenshot collection. Five project PNG files reproduced as Figures 1 to 5.